

Exercise #4

6. February 2023

Problem 1. (Bisection Method)

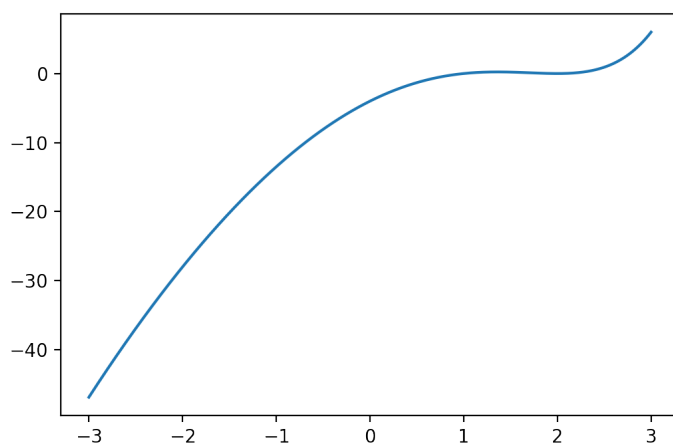
Consider the function

$$f(x) = (1 - x)2^{x+2} + (2^x - 2)x^2 + 8(x - 1),$$

we want to find roots of $f(x)$ on the interval $[-3, 3]$.

- a) Plot the function on the interval $[-3, 3]$.
- b) Do 4 iterations of the bisection method by hand. What root(s) did you find and what is the error bound?
- c) How many iterations are required to guarantee an error smaller than 10^{-3} ?
- d) Use python to find roots of $f(x)$ with an error smaller than 10^{-3} .

Solution.



a)

b) We compute

$$f(-3) \cdot f(3) = -281.25 < 0,$$

and note that $f(x)$ is a continuous function. Hence there is at least one root. Set

$$\begin{aligned} a_0 &= -3, \\ b_0 &= 3, \\ c_0 &= \frac{a_0 + b_0}{2} = 0, \end{aligned}$$

and compute

$$\begin{aligned} f(a_0) \cdot f(c_0) &= 187.5 > 0, \\ f(c_0) \cdot f(b_0) &= -24 < 0, \end{aligned}$$

to see that there is a root in $[0, 3]$. Then:

$$\begin{aligned} a_1 &= 0, \\ b_1 &= 3, \\ c_1 &= \frac{3}{2}, \end{aligned}$$

and

$$f(a_1) \cdot f(c_1) \approx -0.828 < 0,$$

$$f(c_1) \cdot f(b_1) \approx 1.243 > 0,$$

so there is a root in $[0, \frac{3}{2}]$.

$$a_2 = 0,$$

$$b_2 = \frac{3}{2},$$

$$c_2 = \frac{3}{4},$$

$$f(a_2) \cdot f(c_2) \approx 1.989 > 0,$$

$$f(c_2) \cdot f(b_2) \approx -0.103 < 0,$$

so there is a root in $[\frac{3}{4}, \frac{3}{2}]$.

$$a_3 = \frac{3}{4},$$

$$b_3 = \frac{9}{8},$$

$$c_3 = \frac{3}{2},$$

$$f(a_3) \cdot f(c_3) \approx -0.07 < 0,$$

$$f(c_3) \cdot f(b_3) \approx 0.029 > 0,$$

so there is a root in $[\frac{6}{8}, \frac{9}{8}]$. This gives the root $c_4 = \frac{15}{16}$ with error bounded by $\frac{3}{16}$.

- c) The error after n iterations is $\frac{6}{2^n} = \frac{3}{2^{n-1}}$, so we have to find the smallest integer n such that

$\frac{3}{2^{n-1}} < 10^{-3}$, which is equivalent to finding the smallest integer n such that $2^{n-1} > 3 \cdot 10^3$.

$$\begin{aligned}2^1 &= 2, \\2^2 &= 4, \\2^3 &= 8, \\2^4 &= 16, \\2^5 &= 32, \\2^6 &= 64, \\2^7 &= 128, \\2^8 &= 256, \\2^9 &= 512, \\2^{10} &= 1024, \\2^{11} &= 2048, \\2^{12} &= 4096 > 3000.\end{aligned}$$

13 iterations are required.

Alternatively use log to solve $\min\{n \in \mathbb{N} : 2^{n-1} \geq 3000\}$.

Problem 2. (Radius of Convergence)

Consider the function

$$f(x) = x \log(x) - x.$$

a) Show that the minimal function $M(a, b)$ such that

$$\left| \frac{f''(y)}{f'(x)} \right| \leq 2M(a, b),$$

for all $x, y \in [a, b]$, where $0 < a < b < \infty$, is given by

$$M(a, b) = \frac{1}{2a \log(a)}, \quad \text{if } a > 1,$$

and that no such function exists if $a \leq 1$.

- b) Use the convergence theorem to show that Newton's method converges for x_0 satisfying both of the two inequalities

$$2(e - |x_0 - e|) \log(e - |x_0 - e|) \geq |x_0 - e|,$$

$$|x_0 - e| < |e - 1|,$$

where e is Euler's number.

- c) Conclude how to find for which x_0 Newton's method is guaranteed to converge for $f(x)$, by looking at the fix points of the function

$$g(x) = 2(e - x) \log(e - x).$$

Can fixed point iterations of this function g be used to find this fixed point?

- d) Use the bisection method to find the fixed point of $g(x)$. For which x_0 is Newton's method guaranteed to converge for $f(x)$?
- e) Verify numerically that Newton's method converges for some x_0 that you found in part d), also test some x_0 that were not found in part d).

Solution.

- a) We compute the derivatives to get the inequality:

$$|f'(x)| = |\log(x)| \geq \begin{cases} \log(a), & \text{if } a > 1, \\ 0, & \text{if } a \leq 1 \end{cases},$$

$$f''(x) = \frac{1}{x} \leq \frac{1}{a}.$$

This gives

$$M(a, b) = \frac{1}{2a \log(a)}, \quad \text{if } a > 1,$$

and no such $M(a, b)$ exists for $a \leq 1$.

- b) It is analytically clear that the only root for $f(x)$ is $f(e) = 0$. The convergence theorem tells us that Newton's method converges if $|x_0 - e| \leq \min\{\frac{1}{M(e-|x_0-e|, e+|x_0-e|)}, |x_0 - e|\}$ and $[e - |x_0 - e|, e + |x_0 - e|] \subset (1, 2e - 1)$. So it converges if $2(e - |x_0 - e|) \log(e - |x_0 - e|) \geq |x_0 - e|$ and $|x_0 - e| < |e - 1|$.

c) The inequality

$$2(e - |x_0 - e|) \log(e - |x_0 - e|) \geq |x_0 - e|$$

is equivalent to

$$2(e - |x_0 - e|) \log(e - |x_0 - e|) - |x_0 - e| \geq 0,$$

so we have to find for which x the function

$$h(x) = 2(e - x) \log(e - x) - x$$

is positive. Note that $h(0) = 2e \geq 0$ and that

$$h'(x) = -2 \log(e - x) - 3$$

is negative for $x \in [0, e - 1]$, hence the Newton method will converge for all $x_0 \in [e - \tilde{x}, e + \tilde{x}]$ where $\tilde{x}$ solves the equation $h(x) = 0$, meaning that $\tilde{x}$ is a fix-point to $g(x)$. The function $g(x)$ has derivative

$$\begin{aligned} |g'(x)| &= |-2 \log(e - x) - 2| \\ &= 2 + 2 \log(e - x) \\ &\geq 1, \end{aligned}$$

for $x \in [0, e - 1]$. Hence fixed-point iterations will not converge for the function $g(x)$.

d) $g(1.21779) \approx 1.21779$, so Newton's method converges for $f(x)$ when $x_0 \in [e - 1.21779, e + 1.21779] \approx [1.50049, 3.93607]$.

e) Using an $x_0 \in [0, 1]$ will give an error as $x_1 \leq 0$ is not in the domain of $f(x)$. Newton's method will converge for all $x_0 \in (1, \infty)$. The convergence theorem gives a sufficient condition for convergence, but not a necessary condition.

Problem 3. (Periodic functions)

Recall that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is called periodic, if there exists $p > 0$ (a period of f) such that

$$f(x + p) = f(x) \text{ for all } x \in \mathbb{R}.$$

Moreover, the smallest positive number for which this statement holds (if it exists), is called the fundamental period of f .

a) Decide whether the following statement is true or false, and then find either a proof or a counterexample: Every periodic function has a fundamental period.

b) What is the fundamental period of the following functions:

- $f(x) = \sin(x + 2)$
- $f(x) = \pi \cos(\pi x)$
- $f(x) = \cos(\frac{2\pi}{m+1}x) + \sin(\frac{2\pi}{n-1}x), \quad n, m \in \mathbb{N} \setminus \{1\}.$

Solution.

a) The statement is false. Counterexample: Consider any constant function $f(x) = c$, for some $c \in \mathbb{R}$. Then, for any $x \in \mathbb{R}$ and $h > 0$, we have that $f(x + h) = c = f(x)$. Hence, we cannot find the smallest positive number such that the previous equation holds, i.e. there is no fundamental period of f .

A more interesting example is given by the function

$$g(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

It is clear that for any positive rational number q , we have that $g(x + q) = g(x)$, for any $x \in \mathbb{R}$. However, we cannot find the minimum positive rational number that satisfies the previous equation.

b)

- $f(x) = \sin(x + 2) = \sin(x) \cos(2) + \sin(2) \cos(x)$. Since $\cos(2)$ and $\sin(2)$ are just constants and $\cos(x)$ and $\sin(x)$ have fundamental period 2π , the fundamental period is 2π .
- $f(x) = \pi \cos(\pi x)$. Recall that the fundamental period of $\cos(x)$ is 2π . Now, if p is the fundamental period of f , then

$$\cos(\pi x) = \cos(\pi(x + p)) = \cos(\pi x + \pi p).$$

Hence the fundamental period of f is $p = 2$.

- $f(x) = \cos(\frac{2\pi}{m+1}x) + \sin(\frac{2\pi}{n-1}x)$, $n, m \in \mathbb{N} \setminus \{1\}$. In the same way that the previous item, we have that the fundamental period of $\cos(\frac{2\pi}{m+1}x)$ is $m+1$, and the fundamental period of $\sin(\frac{2\pi}{n-1}x)$ is $n-1$. Now, f has period p if $\cos(\frac{2\pi}{m+1}x)$ and $\sin(\frac{2\pi}{n-1}x)$ have period equal to p . In other words, for $\cos(\frac{2\pi}{m+1}x)$, we have to find a multiple of $m+1$ which is also a multiple of $n-1$, and similarly for $\sin(\frac{2\pi}{n-1}x)$. Thus, the fundamental period of f is given by the least common multiple of $m+1$ and $n-1$.

Problem 4.

Recall that a function f is even if $f(-x) = f(x)$, and odd if $-f(-x) = f(x)$.

- Show that if $f(x)$ is odd, and $g(x)$ is even, then $h(x) := f(x)g(x)$ is odd.
- Show that if $f(x)$ is odd, then $h(x) := f^2(x)$ is even.
- Show that if $f(x)$ is odd, then for any $L > 0$,

$$\int_{-L}^L f(x)dx = 0.$$

- Show that if $f(x)$ is odd and $g(x)$ is even then $f \circ g$ and $g \circ f$ are both even.
- Show that if $f(x)$ is even, then for any $L > 0$,

$$\int_{-L}^L f(x)dx = 2 \int_0^L f(x)dx.$$

Solution.

- We compute $h(-x) = f(-x)g(-x) = -f(x)g(x) = -h(x)$. So h is odd.
- We compute $h(-x) = (-f(x))^2 = f^2(x) = h(x)$. So h is even.

c) We rewrite the integral,

$$\int_{-L}^L f(x) dx = \int_{-L}^0 f(x) dx + \int_0^L f(x) dx$$

Note that

$$\int_{-L}^0 f(x) dx = - \int_0^{-L} f(x) dx = \int_0^L f(-x) dx = - \int_0^L f(x) dx.$$

Hence, the integral is 0.

d) We compute $h(-x) = f \circ g(-x) = f(g(-x)) = f(g(x)) = h(x)$. So h is even. Similarly $h(-x) = g \circ f(-x) = g(f(-x)) = g(-f(x)) = g(f(x)) = h(x)$. So h is again even.

e) We rewrite the integral,

$$\int_{-L}^L f(x) dx = \int_{-L}^0 f(x) dx + \int_0^L f(x) dx$$

Note that

$$\int_{-L}^0 f(x) dx = - \int_0^{-L} f(x) dx = \int_0^L f(-x) dx = \int_0^L f(x) dx.$$

Hence, the integral is $2 \int_0^L f(x) dx$.

Problem 5. (Trigonometric interpolation)

In our first lectures, we saw how polynomials can be used to interpolate functions or discrete datasets. That is actually not the only possibility, and in fact we can also use trigonometric functions for interpolation. Namely, the set $\{1, \sin x, \cos x, \sin 2x, \cos 2x, \dots\}$ is a common basis for that.

Let $y = f(x) = 2 + \sin(x) \cos(x)$ be the function we wish to interpolate, for which we consider three data points:

i	0	1	2
x_i	0	$\pi/6$	$\pi/3$
y_i	2	$2 + \sqrt{3}/4$	$2 + \sqrt{3}/4$

Our task here is to construct a trigonometric function $p(x)$ that goes through these three points. Since $f(x)$ is just a constant plus an odd function, it suffices to use only sinus terms plus the constant for the interpolation. That is, we are looking for some

$$p(x) = \alpha_0 + \alpha_1 \sin x + \alpha_2 \sin 2x$$

such that $p(x_i) = y_i$ for $i = 0, 1, 2$, with α_0 , α_1 and α_2 being three real coefficients to be determined.

- Based on the data in table, compute the interpolation coefficients α_0 , α_1 and α_2 .
- Show that, for this particular case, we have $p(x) = f(x)$ for all $x \in \mathbb{R}$, that is, the interpolant is identical to the function being interpolated.
Hint: remember the trigonometric identity $\sin 2x = 2 \sin x \cos x$.

Solution.

- a) By enforcing $p(x_i) = y_i$, we get

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1/2 & \sqrt{3}/2 \\ 1 & \sqrt{3}/2 & \sqrt{3}/2 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 + \sqrt{3}/4 \\ 2 + \sqrt{3}/4 \end{pmatrix} \Rightarrow \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ \frac{1}{2} \end{pmatrix}.$$

Therefore, $p(x) = 2 + \frac{1}{2} \sin(2x)$

- b) Since $\sin 2x = 2 \sin x \cos x$, we have that

$$p(x) = 2 + \frac{1}{2} \sin(2x) = 2 + \frac{1}{2} 2 \sin(x) \cos(x) = 2 + \sin(x) \cos(x) = f(x).$$